Tire Scout - Weekly Technology Digest

Week 2026-10-05 to 2026-10-11 · generated 2026-10-12 17:43 IST

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Priority developments

1. Evonik next-generation silane system for silica tire compounds

Organization	Evonik
Category / maturity	Rubber compounds and materials / commercial
Score	7.6/10 - rel 9.0, nov 7.0, ev 6.0, imp 7.0, act 8.0
Confidence	medium
Published	2026-10-06T09:00:00+02:00

What changed: Launch of a new silane system claimed to lower rolling resistance versus Evonik's previous generation.

Technical detail: Improved silane coupling promotes silica dispersion and polymer-filler interaction in the tread compound.

Claimed benefit: 12 percent lower rolling resistance in laboratory compound testing. *Conditions:* Laboratory compound testing versus the company's previous silane generation; abrasion index unchanged.

Limitations: Supplier press release; no independent verification; tire-level results not shown.

Follow-up: Request technical data sheet and sample for internal mixing trial.

Source: https://evonik.com/en/news/press-releases/2026/10/silane-silica-ev.html

2. Bekaert details higher-tensile steel cord for EV tire belts

Organization	Bekaert
Category / maturity	Reinforcement and construction / prototype
Score	6.3/10 - rel 8.0, nov 6.0, ev 4.0, imp 6.0, act 6.0
Confidence	medium
Published	2026-10-05T10:00:00+00:00

What changed: Higher-tensile cord grade presented for EV belt constructions.

Technical detail: Higher tensile strength allows thinner cords and lower belt mass.

Claimed benefit: Belt mass reduction while maintaining stiffness. *Conditions:* Comparison versus supertensile grade at equal safety factor.

Limitations: Trade coverage of a supplier presentation; no independent test data. [access: partial]

Follow-up: Ask Bekaert for cord datasheet and gauge/fatigue data.

Source: http://tirebusiness.com/article/steel-cord

3. HF Mixing Group tandem mixing energy savings technical paper

Organization	HF Mixing Group
Category / maturity	Manufacturing and process / commercial
Score	5.8/10 - rel 7.0, nov 5.0, ev 5.0, imp 5.0, act 6.0
Confidence	medium
Published	unknown

What changed: Technical paper quantifies energy savings of tandem mixing configuration.

Technical detail: Two-stage tandem mixing reduces peak power and improves temperature control.

Claimed benefit: Up to 15 percent specific energy reduction in the mixing room. *Conditions:* Case data from one reference line; compound dependent.

Limitations: Supplier case study; single-line data. [access: partial]

Follow-up: Benchmark against internal mixing energy baselines.

Source: https://hf-mixinggroup.com/downloads/tandem-mixing-2026.pdf

Developing themes

• **Filler-interface chemistry keeps advancing on rolling resistance** (1 development(s))

• **EV-driven lightweighting across reinforcement and process** (2 development(s))

Practical team actions

1. Request the Evonik silane technical data sheet and plan a lab mixing trial
2. Ask Bekaert for the mega-tensile cord datasheet and fatigue data
3. Benchmark tandem-mixing energy figures against internal mixing-room baselines

Appendix: index by daily edition

Date	Status	Developments
2026-10-05	research_complete	Evonik next-generation silane system for silica tire compounds; Review of silica-silane coupling chemistry in tire tread compounds; Bekaert details higher-tensile steel cord for EV tire belts; Blog claims devulcanization process with 90 percent property retention; HF Mixing Group tandem mixing energy savings technical paper
2026-10-06	no_updates	-
2026-10-07	no_updates	-
2026-10-08	no_updates	-
2026-10-09	no_updates	-
2026-10-10	no_updates	-
2026-10-11	no_updates	-